Course name: Computer networks 1
Academic year: 2025-2026

Ինֆորմատիկայի բակալավր (հայկական)/Licence en Informatique de l'Université de Toulouse (ֆրանսիական)		
Qualification : Computer Science		
Coordinating faculty / chair : IMA		
Year of study : 3 Semester : 1		3.0 ECTS
Total hours: 30.0	Lecture (CM): 18.0 Tutorials (TD) : Lecture and Tutorials CTD : Practicals (TP) : 12.0 Lecture and Practicals (CTP): Project:	
Estimated number of hours of individual student work (TPS): 60.0		
Language of instruction : English		
Instructor (CM) : Tadevosyan Robert Instructor (TP) : Tadevosyan Robert		
Instructor's email : Robert.Tadevosyan@pers.ufar.am		

PURPOSE AND DESCRIPTION OF THE COURSE

This course will cover the principles and working processes of computer networks. Will be revealed the network structure, starting from the lowest physical layer till application layer, and work in the Internet. Will be studied the principles of work of the network protocols as well as the ways of diagnosing the problems. Also will be the reviewed the mainly used networking equipment and how monitor network configuration examples of commonly encountered problem. Also will be the lab works, based on existing equipment, for getting the practical knowledge of creating networks, based on theoretical studies of the studied material.

INTENDED LEARNING OUTCOMES

Knowledge and skills

A - Knowledge	A1.1. Describe the operation of a computer and its environment. (N2) Niveau d'Application A1.2. Analyze the fundamental components and architectural models (e.g., client-server, peer-to-peer, layered architectures like TCP/IP and OSI) that comprise modern computer networks. (N2) Niveau d'Application
	A1.3. Explain the purpose, functions, and interactions of the various layers within the OSI and TCP/IP models, including the key protocols operating at each layer and their impact on network communication. (N2) Niveau d'Application
B1 – Skills to apply professional knowledge	B1.1. Study computer networks. (N2) Niveau d'Application B1.2. Configure basic network settings (e.g., IP addressing, subnet masks, default gateways) on common operating systems and network devices (e.g., switches, routers), demonstrating foundational connectivity. (N2) Niveau d'Application B1.3. Utilize network monitoring tools (e.g., ping, traceroute, Wireshark) to capture and analyze network traffic, identifying common network problems and verifying protocol behavior at different layers. (N2) Niveau d'Application B1.4. Implement simple client-server applications using basic network programming interfaces (e.g., sockets), demonstrating an understanding of how applications interact with the network stack. (N2) Niveau d'Application
B2 – General (transversal) skills	B2.1. Utilize professional literature in English., (N2) Niveau d'Application B2.2. Apply written and oral professional communication in English., (N2) Niveau d'Application B2.3. Collaborate effectively in teams to troubleshoot and resolve simulated network connectivity issues, communicating technical concepts clearly to both technical and non-technical audiences., (N2) Niveau d'Application B2.6. Apply systematic problem-solving methodologies to analyze and diagnose complex technical issues, drawing parallels between network troubleshooting techniques and problem-solving in other domains., (N2) Niveau d'Application

KNOWLEDGE / SKILLS ASSESSMENT & EVALUATION			
Form of assessment	**Ongoing evaluation** ☒	**Midterm exam** ☐	**Final exam** ☒
Exam type	**Oral** ☒ **Written** ☒	**Oral** ☐ **Written** ☐	**Oral** ☐ **Written** ☒

Exam task	**Practice part (20 points)** Students will be evaluated by their activity on the practical parts of the course. **Two tasks (10 points each):** The students demonstrates a practice knowledge for design and create the VLAN based network structure and distribute the IP addresses to subnets	e.g., MCQs (**multiple choice questions**), short-answer test, other (please specify)	**Section 1: Knowledge and Theoretical Base Identification (6 points)** This section will focus on your knowledge and theoretical base. It will test your understanding of the core network working principles, including OSI model, wired and wireless connections, LAN physical topology and IP addressing system. You will be required to describe how to crate and design of a network topology and IP addressing. **Section 2: Network Structure Determining (7 points)** This section will check your ability to determine the correct option from several, which will demonstrate your ability to ponder and make the best choice, as well as your practical knowledge. **Section 3: Network Analysis and IP Calculations (6 points)** This section will demonstrate your practical knowledge of creating IP subnetworks. You will need to calculate and present the full Info of IP subnets: IP, Mask. Your response should demonstrate a clear understanding of IP system and technical and conceptual aspects of the topic.
Group work	**yes** ☐ **no** ☒	**yes** ☐ **no** ☐	**yes** ☐ **no** ☒
Duration: (hours)	**0.5**		**2**

Weight in final grade (100%)	25%		75%
Assessment Criteria:	**Full Credit (10 points for each task):** The student provides a completely accurate and well-explained knowledge about core networking principles, including correct IP calculations and a physical/logical network design. **Partial Credit (9-4 points for each task):** The student		**Section 1: Knowledge and Theoretical Base Identification (6 points)** **Full Credit (6 points):** The student provides a completely accurate and well-explained knowledge about core networking principles, including correct IP calculations and a physical/logical network design.

	demonstrates a clear understanding of the core concepts but makes minor errors in execution (e.g., a small mistake in a description or a missing some parts in topic). The score will depend on the severity and number of errors. **Minimal Credit (4-1 points for each task):** The student attempts to present the technology but shows a significant conceptual misunderstanding.		**Partial Credit (5-3 points):** The student demonstrates a clear understanding of the core concepts but makes minor errors in execution (e.g., a small mistake in a description or a missing some parts in topic). The score will depend on the severity and number of errors. **Minimal Credit (2-1 points):** The student attempts to present the technology but shows a significant conceptual misunderstanding. **Section 2: Network Structure Determining (7 points)** **Credit (7-0 points):** The score will depend on the number of right answers. **Section 3: Network Analysis and IP Calculations (6 points)** **Full Credit (6 points):** The student provides a comprehensive, wellstructured, and technically accurate information about of creating IP subnetworks. Their argument is cohesive and demonstrates a deep understanding of the topic and network design. **Partial Credit (5-3 points):** Student shows a general understanding of the topic but may lack technical detail, contain minor inaccuracies, or have a less cohesive structure. The student might correctly identify a concept but fail to elaborate on its significance.

			Minimal Credit (2-1 points): The student provides a very brief or poorly response that fails to make a clear and accurate connection between the topics. The answer may be largely irrelevant to the prompt.

TEACHING METHODS AND TOOLS

Lecture, Explanation, Slideshow, Presentation, Demonstration, Video-presentation, Practical Work, Group work, Debate, Exercises, Instruction with demonstration, Problem-solving learning

KNOWLEDGE & SKILLS PREREQUISITS

Basic Computer Skills	It is important to have a good understanding of how to use computers, including knowledge of operating systems like Windows, macOS, or Linux.
Fundamental Knowledge of Computer Science	Concepts such as data structures, algorithms, and programming languages are beneficial as they form the basis of understanding how network protocols work.
Understanding of the Internet	Familiarity with how the internet works, including concepts like IP addresses, domain names, and how data is transmitted over the internet, can be helpful
Knowledge of Operating Systems	Understanding how operating systems manage resources, handle processes, and provide networking services is essential for grasping networking concepts

Mathematics	Basic knowledge of mathematics, including algebra and probability, can be useful for understanding certain networking concepts like error detection and correction. Good understanding of mathematical concepts such as binary and hexadecimal numbering systems help in understanding computer networking
Programming Skills	While not always a strict requirement, having programming skills in languages like Python, Java, or C can be beneficial for tasks such as network programming and automation
Curiosity and Problem-Solving Skills	Networking often involves complex problem-solving, troubleshooting, and critical thinking. Having a curious mindset and the ability to troubleshoot issues systematically are important skills in this field
Resources	It's helpful to have access to resources such as textbooks, online courses, tutorials, and networking equipment (like routers and switches) for hands-on practice

COURSE SYLLABUS AND RESOURCES

Topic	Hours	Core resources	Additional resources
CM			
1. Introduction. What is computer network 1.1. Computer networks; PAN, LAN, WAN	CM 1,5h	Title, page	Title, page

2. OSI Model 2.1. Channel and packet switching 2.2. Protocol description 2.3. OSI model 2.4. Physical Layer 2.4.1. Cooper - Coaxial, Twisted Pair (UTP, STP, FTP), 2.4.2. Fiber Optics (SM, MM), 2.4.3. Wireless systems; Wi-Fi standards 2.4.4. Satellite	CM 7.5h TD 1.5h		
3. Data Link Layer 3.1. Network topologies (Bus, Ring, Star, Mesh) 3.2. Token Ring, Ethernet	CM 3h TD 3h		

technologies - working technologies and compares. 3.3. Ethernet CSMA/CD technology. 3.4. Ethernet networks - problems and solutions; 3.5. Hub and Switching technology difference 3.6. Smart Switch functionality: VLAN technology			

4. Network Layer 4.1. NETBIOS/NETBEUI, IPX/SPX, TCP/IP Protocols - working principals and difference 4.2. Boolean and decimal systems 4.3. IP addressing and classes; Private and Public addresses 4.4. Network Mask 4.5. ICMP protocol: Ping, Traceroute Tools	CM 6h TD 7.5h		
TD			
1. Physical and Data Link Layer 1.1. UTP cabling 1.1.1. UTP Cable stripping and RJ45connecting 1.2. MikroTik devices first look	TD 4.5h	Title, page	Title, page
2. Work with IP addresses 2.1.1. IP address calculations (IP, masks and subnets) 2.2. Configuring IP address in	TD 7.5h		
OS/s and testing. 2.3. Working with MikroTik devices; Initial configuring, updating.			

CORE BIBLIOGRAPHY

Core textbooks, publications, and research articles:

1. Pearson, Computer Networking: A Top-Down Approach (8th Edition) (2021)

2. Tilted Windmill Press, Networking for Systems Administrators (IT Mastery) (2019)

3. Cisco Press, Routing TCP/IP, Volume II: CCIE Professional Development (2nd Edition) (2016)
4. O'Reilly Media, Network Warrior (2nd Edition) (2011)
5. Wiley, The All-New Switch Book: The Complete Guide to LAN Switching Technology (2nd Edition) (2008)
6. Cisco Press, Routing TCP/IP, Volume 1 (2nd Edition) (2005)
7. Networking with MikroTik RouterOS: A Practical Approach to Understanding and Implementing RouterOS (updated to RouterOS v7). MTCNA Certification Course
8. MikroTik Router OS by Example book, 2nd Edition, Color (2017)

Supplementary textbooks, publications, and research articles:

1. Practical Packet Analysis, 3E: Using Wireshark to Solve Real-World Network Problems 3rd Edition (2017)
2. Computer Networking with Internet Protocols and Technology/Authors: William Stallings/Pearson
3. Paul Love, Joe Merlino, Jeremy C./Beginning Unix/Wiley Publishing, Inc
4. Joseph Davies and Tomas Lee/Microsoft Windows Server2003 TCP/IP protocols
5. And Services Technical Reference/Microsoft Press, 2003, ISBN : 0-73-5612919
6. Mattew J. Castelli/Lan Switching first-step/ Cisco Press, 2004, ISBN : 1-58720-100-3
7. David Barnes, Basir Sakandar/Cisco LAN Switching Fundamentals/Cisco Press, 2004, ISBN 1-58705-089-7

Web resources :

1. http://www.petri.co.il
2. http://technet.microsoft.com
3. http://www.cisco.com/
4. https://en.wikipedia.org/wiki/Computer_network
5. https://www.journals.elsevier.com/computer-networks/
6. http://www.cs.vu.nl/~ast/